Case Study 2: Examination Result and Grace-Mark Validation

1. Problem Statement

Read marks for multiple subjects, check pass criteria, optionally apply grace marks to a subject that is short by one mark, and display the final result without exceeding maximum marks.

2. Objectives

Use lists and loops to process subject marks.

Apply subject-wise pass criteria.

Validate marks and display pass/fail status.

3. Assumptions

There are 3 subjects, each out of 100.

A student needs at least 35 marks in every subject to pass.

If a subject has exactly 34 marks, one grace mark may be added, but only once overall.

Marks must be between 0 and 100.

4. Algorithm

1. Start.

2. Input marks for three subjects and validate each mark.

3. Store the marks in a list.

4. If grace has not been used and a mark is 34, add one mark and mark grace as used.

5. Check whether every final subject mark is at least 35.

6. Display original/final marks and the result.

7. Stop.

5. Flowchart (text form)

6. Pseudocode

START

marks = empty list; grace_used = FALSE

REPEAT for 3 subjects: INPUT mark; validate 0 <= mark <= 100; append mark

FOR each mark: IF mark == 34 AND grace_used == FALSE, add 1 and set grace_used = TRUE IF every mark >= 35 THEN result = PASS ELSE result = FAIL

DISPLAY final marks and result

STOP

7. Complete Python Source Code

```
marks = []

for i in range(3):
mark = int(input(f"Enter marks for subject {i + 1} (0-100): "))

while mark < 0 or mark > 100:
print("Invalid marks. Enter a value from 0 to 100.")
mark = int(input(f"Re-enter marks for subject {i + 1}: "))
marks.append(mark)
```

```
original_marks = marks.copy()
grace_used = False
    for i in range(len(marks)):
     if marks[i] == 34 and not grace_used:
marks[i] = marks[i] + 1
grace_used = True
    print("
    Original marks:", original_marks)
    print("Final marks:", marks)
    passed = True
    for mark in marks:
     if mark < 35:
     passed = False
    if passed:
     print("Result: PASS")
    else:
     print("Result: FAIL")
    if grace_used:
     print("Grace mark used: Yes")
    else:
     print("Grace mark used: No")
```

8. Test Case and Expected Output

Inputs: 34, 67, 80.

Original marks: [34, 67, 80] Final marks: [35, 67, 80] Result: PASS Grace mark used: Yes

9. Conclusion

The program validates marks, applies the stated one-mark grace policy, and determines the final result.